$x_m = 5.3852 \times 10^{-2}\,m \cong 5.39$ cm

Divide term by term:

$$\left.\begin{array}{l}\sin\varphi > 0 \\ \text{and} \\ \cos\varphi < 0\end{array}\right| \quad tg\varphi = -0.4 \quad \Rightarrow \varphi \cong (158.2)^\circ \text{ or } \varphi \cong (2.76) \text{ rd}$$

$x_{(cm)} \cong 5.39 \sin(4t + 2.76)$

$\Rightarrow V_{(cm/s)} \cong 21.6 \cos(4t + 2.76)$

Question 4.2

Solve application 4.2 if $v_0 = +\,0.20$ m/s, and all other data are unchanged.

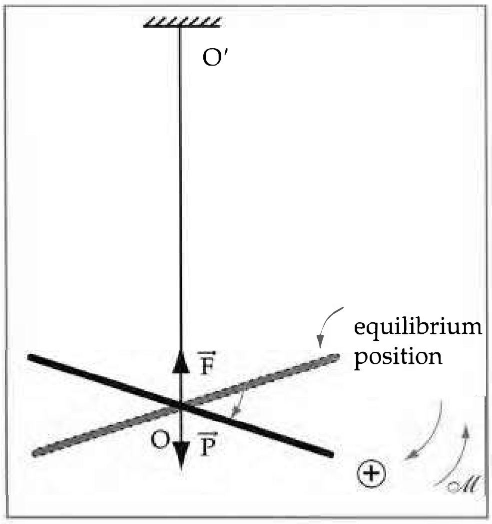

Figure 4.11.a

b) Torsion pendulum

i) <u>Torsion of a cylindrical wire</u>

One extremity of a vertical steel wire is connected to a fixed support. Its other extremity is connected to the midpoint of a horizontal rod (Figure. 4.11-a).

The rod is rotated about the wire, in a horizontal plane, through a small angle θ ; thus the wire is twisted through the same angle. The wire then exerts a restoring torsion couple whose moment (w. r. t. OO') (Figure 4.11-b) is such that:

$\mathcal{M} = -\,C\theta$

• $\theta = 0$ corresponds to the equilibrium position.

This formula is analogous to the restoring force exerted by a spring where C is the torsion constant of the wire which depends on its length, cross sectional area, and its mechanical nature.

θ is measured in radians but the radian does not appear in the equation and C are measured in m.N.

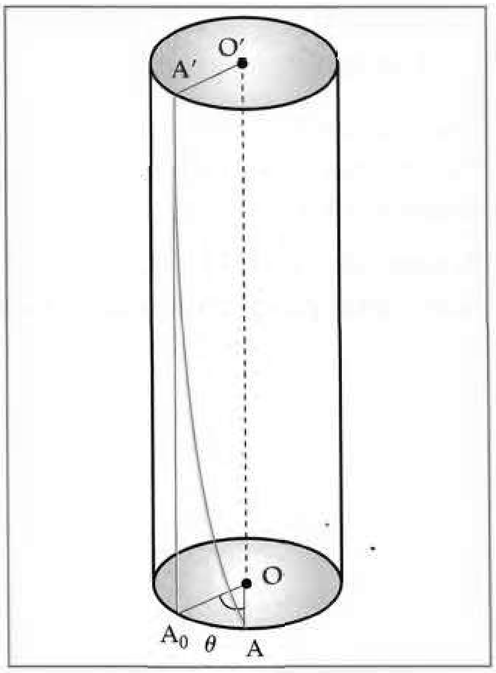

Figure 4.11.b

ii) **Elastic potential energy of a cylindrical torsion wire:**

When the wire is twisted by a small angle θ_m ($\leq 10^o$), the torsion potential energy of the wire is E_P. In order to twist the wire through an angle θ, one must apply a couple of forces whose moment is $\mathcal{M}' = -\mathcal{M} = C\theta$.

$E_P = W$, where W is the work done by $\mathcal{M}'$ on the wire to twist it slowly from 0 to θ.

For rotation $d\theta$, starting from θ (figure 4.12), over which $\mathcal{M}'$ is supposed constant, the work done is:

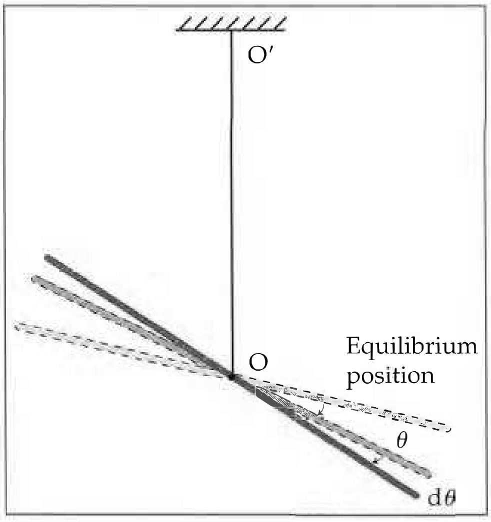

Figure 4.12

$dW = \mathcal{M}'.d\theta = C.\theta.d\theta \quad \Rightarrow \quad \frac{dW}{d\theta} = C.\theta$

This equality represents the derivative of W with respect to θ. Find the integral with respect to θ:

$$W = \frac{1}{2}C.\theta^2 + cte$$

For $\theta = 0$ rd. $\Rightarrow W_0 = cte = 0$

For any angle θ $W = \frac{1}{2}C.\theta^2$

$$E_{P_\theta} = W = \frac{1}{2}C.\theta^2$$

For the angle θ_m: $E_P = \frac{1}{2}C.\theta_m^2$

iii) **Torsion Pendulum**

The chosen system is (torsion wire, rod, Earth, support). I is the moment of inertia of the rod with respect to the axis of rotation OO'.

Rotate the rod by $\theta_m \leq 10°$ about the axis OO' in the horizontal plane (the wire is twisted by θ_m) and release it at instant $t_0 = 0$ ($\theta'_0 = 0$ and $\theta_0 = \theta_m$).

At $t_0 = 0$:

$[(E_K)_0 = 0 \quad ; \quad E_{P_{torsion}} = \frac{1}{2}C\theta_m^2 \quad ; \quad E_{P_{Grav}} = E_{PG}]$

The rod starts oscillating about its equilibrium position.

At any instant t: The rod has an angular displacement θ and angular velocity θ' (Figure 4.11)

$[E_K = \frac{1}{2}I\theta'^2 \quad ; \quad E_{P_{torsion}} = \frac{1}{2}C\theta^2 \quad ; \quad E_{P_{Grav}} = E_{PG}]$

The mechanical energy of the system is conserved

$E_M = E_K + E_{P_{torsion}} + E_{PGrav} = cte\ \forall t$

$E_M(\theta_m) = E_M(\theta)$

$$\frac{1}{2}C\theta_m^2 + E_{PG} = \frac{1}{2}I\theta'^2 + \frac{1}{2}C\theta^2 + E_{PG}$$

We have the same gravitational potential energy since the rod is always horizontal.

Assume that the horizontal plane of the rod is the zero level of the gravitational potential energy $E_{PG} = 0$.

$$E_M = \frac{1}{2}I\theta'^2 + \frac{1}{2}C\theta^2 = \frac{1}{2}C\theta_m^2 = cst$$

Derive with respect to time:

$I\theta'\theta'' + C\theta\theta' = 0 \quad ; \quad \theta'(I\theta'' + C\theta) = 0 \quad$ with $\theta' \neq 0$

$\Rightarrow I\theta'' + C\theta = 0 \quad$ or $\quad \theta'' + \frac{C}{I}.\ \theta = 0$

This is a second order differential equation of the form:

$\theta'' + \omega_0^2\ \theta = 0 \quad$ where $\quad \omega_0^2 = \frac{C}{I} \quad$ and $\quad \omega_0 = \sqrt{\frac{C}{I}}$

It governs the motion of the torsion pendulum, and its solution is

$\theta = \theta_m \sin(\omega_0 t + \varphi)$

called the time equation of motion.

θ and θ_m have the same unit.

θ is the angular abscissa or the angular elongation.

$\theta_m = cst > 0$; θ_m is the angular amplitude or the maximum angular elongation.

$\omega_0 = cst > 0$; ω_0 is the angular frequency (in rd/s)

$\omega_0 t + \varphi$ is the phase at any time t (in rd)

$\varphi = cte$; φ is the initial phase at $t_0 = 0$ (in rd)

$f_0 = \frac{\omega_0}{2\pi}$ is the proper frequency (in Hz)

$T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$ is the proper period (in s). $\quad T_0 = 2\pi\sqrt{\frac{I}{C}}$

Application 4.3

A rod is of mass $m = 40.0 \times 10^{-2}$ kg and of length $l = 36.0$ cm. The torsion constant of the wire is $C = 0.108$ m.N. The moment of inertia of the rod with respect to the axis of rotation is $I = \frac{ml^2}{12}$. Determine one characteristic of the motion of the oscillator.

■ Solution

The characteristics of the oscillator are ω_0, f_0 and T_0. Choose ω_0.

The proper angular frequecy is: $\omega_0 = \sqrt{\frac{C}{I}}$ with

$$I = \frac{ml^2}{12} = \frac{0.4 \times (0.36)^2}{12} = 4.32 \times 10^{-3} \text{kg.m}^2$$

$$\omega_0 = \sqrt{\frac{0.108}{4.32 \times 10^{-3}}} = 5 \text{rd/s}$$

(The proper frequency is: $f_0 = \frac{\omega_0}{2\pi} = \frac{5}{2 \times 3.14} = 0.795$ Hz)

Application 4.4

Consider the torsion pendulum of application 4.3.

a) Determine the torsion potential energy of the wire for $\theta_m = 0.1$ rd.

b) How are the torsion potential energy and the kinetic energy of the pendulum distributed for $\theta = \frac{\theta_m}{2}$?

■ Solution

a) For θ_m

$$E_{P_{torsion}} = \frac{1}{2} C\theta_m^2$$

$E_{P_{torsion}} = \frac{1}{2} \times 0.108 \times (0.1)^2 = 5.40 \times 10^{-4}$ J

b) Since the mechanical energy of the system (wire, rod, support, Earth) is conserved, therefore:

$$E_M = \frac{1}{2} C\theta_m^2 = \frac{1}{2} C\theta^2 + \frac{1}{2} I\theta'^2$$

$$E_{P_{torsion}} = \frac{1}{2} C\theta^2 = \frac{1}{2} C\left(\frac{\theta_m}{2}\right)^2 = \frac{1}{4} \times 5.4 \times 10^{-4} = 1.35 \times 10^{-4} \text{ J}$$

$$E_K = \frac{1}{2} I\theta'^2 = \frac{1}{2} C\theta_m^2 - \frac{1}{4} \times \frac{1}{2} C\theta_m^2 = \frac{3}{4}\left(\frac{1}{2} C\theta_m^2\right) = 3E_{P_{torsion}}$$

$= 4.05 \times 10^{-4}$ J

c) The compound pendulum:

A compound pendulum is a solid that can oscillate about a fixed horizontal axis under the action of its own weight.

In its stable equilibrium position, the pendulum is under the action of two forces (Figure 4.13). These are:

- Its weight $\vec{W} = m\vec{g}$, vertically downward.
- The reaction force $\vec{R}$ at the axis.

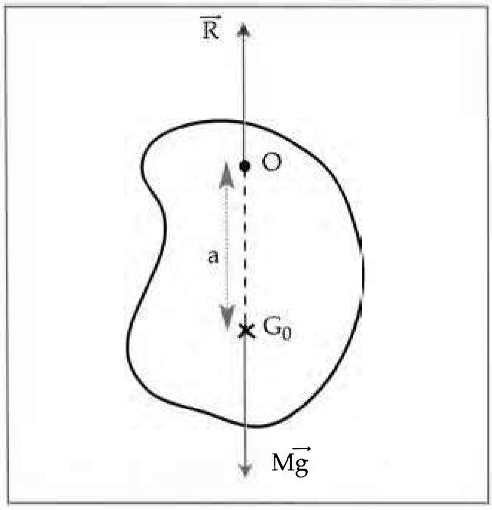

Figure 4.13

Let the zero level of gravitational potential energy be the horizontal passing through the equilibrium position of the center of mass, therefore $E_{P_{Grav.(G0)}} = 0$

Shift the pendulum by $\theta_m \leq 10°$ about the horizontal axis from its stable equilibrium position and release it at $t_0 = 0$. $(\theta'_0 = 0)$.

At $t_0 = 0$:

$(E_k)_0 = 0$; $E_{P_{Grav}} = Mg.G_0H_1$ (Figure 4.14)

$G_0H_1 = G_0O - H_1O = a - a\cos\theta_m = a(1 - \cos\theta_m)$

$E_{P_{Grav}} = Mga.(1 - \cos\theta_m) \cong \frac{1}{2} Mga.\theta_m^2$

$\vec{R}$ has no effect on the motion of the pendulum.

$\Rightarrow E_M \text{ (at } G_1) \cong \frac{1}{2} Mga.\theta_m^2$

The pendulum starts oscillating about its stable equilibrium position.

Remark

For $\alpha \leq 10°$, the approximate value of its cosine is:

$\cos\alpha \cong 1 - \frac{\alpha^2}{2}$ *; where α is measured in rd.*

At any instant t: The angular abscissa is θ and the angular velocity of the pendulum is θ'. (Figure 4.14)

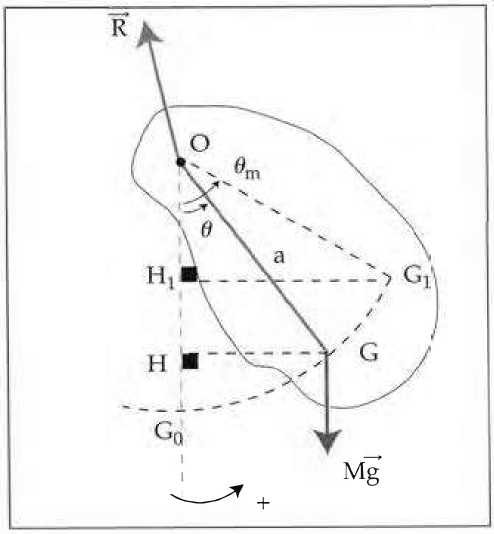

Figure 4.14

I is the moment of inertia of the pendulum with respect to the axis of rotation.

But : $G_0H = G_0O - HO = a - a\cos\theta$

$\Rightarrow E_{P_{Grav}} = Mga.(1 - \cos\theta) \cong \frac{1}{2} Mga.\theta^2$

The mechanical energy of the system (pendulum, support, Earth) is conserved.

E_M (at G_1) = E_M (at G)

$E_M = \frac{1}{2} Mga.\theta_m^2 = \frac{1}{2} I\theta'^2 + \frac{1}{2} Mga.\theta^2 = cte$

Derive with respect to time:

$0 = I\theta'\theta'' + Mga.\theta\theta'$

with $\theta' \neq 0$ during motion.

$$I\theta'' + Mga.\theta = 0 \quad \text{or} \quad \theta'' + \frac{Mga}{I}\theta = 0$$

This is a second order differential equation of the form:

$\theta'' + \omega_0^2\theta = 0$ where $\omega_0^2 = \frac{Mga}{I}$; $\omega_0 = \sqrt{\frac{Mga}{I}}$

It governs the motion of the compound pendulum, and its solution is:

$\theta = \theta_m.\sin(\omega_0 t + \varphi)$

The physical quantities have the same definitions as in the preceding paragraph.

$$T_0 = 2\pi\sqrt{\frac{I}{Mga}}$$

Case of a simple pendulum

Definition

A simple pendulum is formed of a material particle of mass m connected to the free end of an inextensible string of length L and of negligible mass. It can oscillate about a horizontal axis in a vertical plane (Figure 4.15). Since the simple pendulum is a special case of the compound pendulum, therefore all the results obtained for the compound pendulum are applied here with

$I = mL^2$, $a = L$ and $M = m$

Since $\theta_m \leq 10°$ $\theta'' + \left(\frac{g}{L}\right)\theta = 0$; $\omega_0 = \sqrt{\frac{mgL}{ml^2}} = \sqrt{\frac{g}{L}}$

$$\Rightarrow T_0 = 2\pi\sqrt{\frac{L}{g}}$$

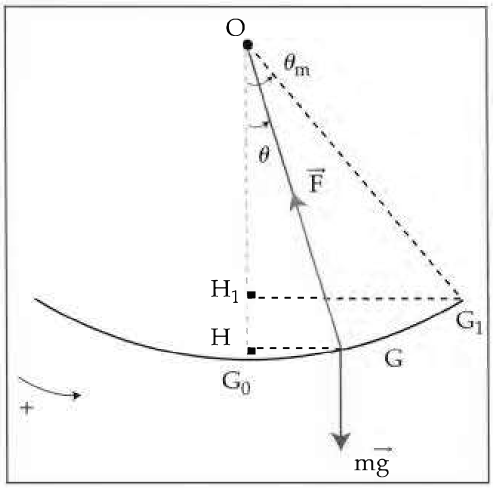

Figure 4.15

The period of a simple pendulum, oscillating with small amplitudes, is directly proportional to the square root of its length and inversely proportional to the square root of the gravitational acceleration at the given location.

Application 4.5

A homogeneous rod of length L = 2 m and of negligible mass can rotate about a horizontal axis perpendicular to the rod and passing through its center O.

Two particles, A and B each of mass m = 0.1 Kg, are fixed at the rod on opposite sides of the axis such that OA = 1 m and OB = 0.5 m (Figure 4.16).

Calculate the proper period of this oscillator (g = 10 m/s²).

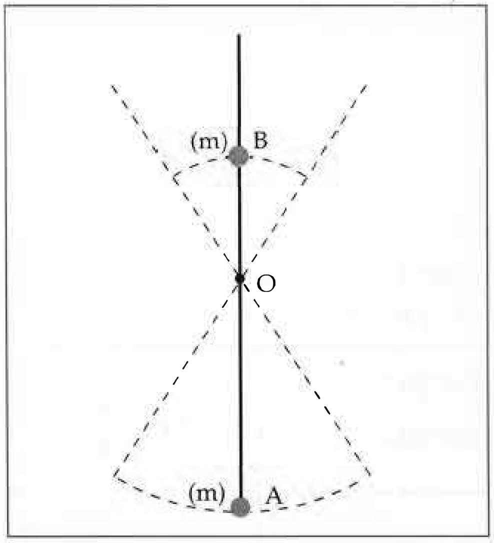

Figure 4.16

■ Solution

- The moment of inertia of the system (rod + m + m) is:

$I = m.\overline{OA}^2 + m.\overline{OB}^2$ since the rod is of negligible mass.

$$I = m\left(\frac{L}{2}\right)^2 + m\left(\frac{L}{4}\right)^2 = \frac{5}{16}mL^2 = \frac{5}{16} \times 0.1 \times 4$$

$I = 0.125$ Kg.m²

- The mass of the system: M = m + m = 0.2 Kg.

- he center of gravity of the system is G such that

$$M.\overrightarrow{OG} = m.\overrightarrow{OA} + m.\overrightarrow{OB}$$

with: $\overrightarrow{OA} = \frac{L}{2}.\vec{i}$ and $\overrightarrow{OB} = -\frac{L}{4}.\vec{i}$

$M.\overrightarrow{OG} = \frac{mL}{2}.\vec{i} - \frac{mL}{4}.\vec{i} = \frac{mL}{4}.\vec{i}$

$\overrightarrow{OG} = \frac{L}{8}\ \vec{i} = 0.25\ \vec{i}$

with $\overrightarrow{OG} = a.\vec{i}$ $\Rightarrow$ $a = 0.25$ m

The proper period is $T_0 = 2\pi\sqrt{\frac{I}{Mga}}$

$= 2\pi\sqrt{\frac{0.125}{0.2\times10\times0.25}} = \pi\ s = 3.14\ s$

	Horizontal elastic pendulum	Torsion pendulum $\theta_m \leq 10°$	Compound pendulum $\theta_m \leq 10°$	Simple pendulum $\theta_m \leq 10°$	General form
Mechanical energy	$v'=x''$; $v=x'$ $\frac{1}{2}mv^2+\frac{1}{2}kx^2=cte$	$\frac{1}{2}I\theta'^2 + \frac{1}{2}C\theta^2 = cte$	$\frac{1}{2}I\theta'^2 + \frac{1}{2}Mga\theta^2 = cte$	$I = mL^2$ $\frac{1}{2}I\theta'^2 + \frac{1}{2}mgL\theta^2 = cte$	$y'^2 + \omega_0^2 y^2 = cte$
Differential equation	$x'' + \frac{k}{m}x = 0$	$\theta'' + \frac{C}{I}\theta = 0$	$\theta'' + \frac{Mga}{I}\theta = 0$	$\theta'' + \frac{g}{L}\theta = 0$	$y'' + \omega_0^2 y = 0$
Proper angular frequency	$\omega_0 = \sqrt{\frac{k}{m}}$	$\omega_0 = \sqrt{\frac{C}{I}}$	$\omega_0 = \sqrt{\frac{Mga}{I}}$	$\omega_0 = \sqrt{\frac{g}{L}}$	ω_0
Proper period	$T_0 = 2\pi\sqrt{\frac{m}{k}}$	$T_0 = 2\pi\sqrt{\frac{I}{C}}$	$T_0 = 2\pi\sqrt{\frac{I}{Mga}}$	$T_0 = 2\pi\sqrt{\frac{L}{g}}$	$T_0 = \frac{2\pi}{\omega_0}$
Solution	$x = x_m\sin(\omega_0 t+\varphi)$	$\theta = \theta_m\sin(\omega_0 t+\varphi)$	$\theta = \theta_m\sin(\omega_0 t+\varphi)$	$\theta = \theta_m\sin(\omega_0 t+\varphi)$	$y = y_m\sin(\omega_0 t+\varphi)$

Table 4.1

The differential equation is of the form

$$y'' + \omega_0^2.y = 0$$

It governs the simple harmonic motion.

The solution of such equation is a sine function of the form:

$y = y_m\sin(\omega_0 t + \varphi)$

y_m is the amplitude of oscillations.

Note that the solution could also be written in the form:

$y = y_m\cos(\omega_0 t+\varphi)$

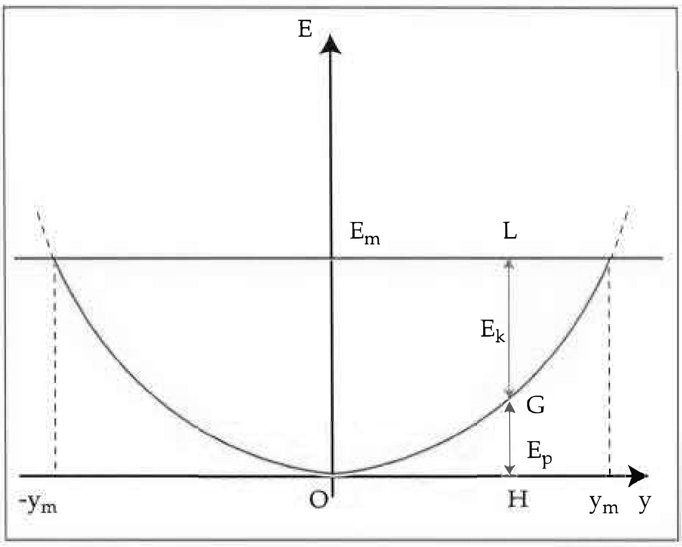

Figure 4.17

Graphical representation:

The mechanical energy of an oscillator performing a simple harmonic motion is:

$E_M = E_K + E_P = cte\ \forall t$

where $E_K = \frac{1}{2}ay'^2$; $E_P = \frac{1}{2}by^2$ and

$E_M = \frac{1}{2}by_m^2$

where y_m is the amplitude of this motion.

The simplest expression is that of E_M which is a constant.

E_M is represented by a straight line parallel to the y'oy axis.

E_P has the form of a parabola.

And $E_K = E_M - E_P$.

The representative curve is shown in figure 4.17.

Take a point G on the graph. GH = E_P ; LH = E_M, and E_K = LG = $E_M - E_P$

4.4 DRIVEN OSCILLATIONS

We saw that the free oscillations of any pendulum (elastic, torsion, or compound) are damped as time passes and eventually disappear. In fact, this happens because of the inevitable frictional forces, whose work causes a decrease in the amplitude of oscillations and therefore in the mechanical energy.

An oscillator is said to be driven when it can perform oscillations with the same amplitude. In order to drive an oscillator, we must provide it with the necessary energy needed to compensate the above-mentioned losses.

How do such oscillators receive energy?

a) Self-driven oscillators

They receive quantities of energy in step with their period.